

Resisted motion in a straight line — Solutions

Warm-up

Question 1:

The net force F_{net} is the vector sum of all forces acting on the object. Since the object is moving in a straight line, we define the direction of motion as positive.

For part (a), the only force is air resistance, which opposes motion:

$$F_{net} = -mkv$$

For part (b), the air resistance is proportional to the square of the velocity:

$$F_{net} = -mkv^2$$

For part (c), there is both air resistance and a constant frictional force:

$$F_{net} = -mkv - 5$$

For part (d), there are two resistive forces (linear and quadratic) and a constant frictional force:

$$F_{net} = -mkv - mkv^2 - 100$$

Question 2:

Using Newton's Second Law, $F_{net} = m\ddot{x}$. Given $m = 5$ kg, we have $\ddot{x} = \frac{F_{net}}{5}$.

For (a):

$$\begin{aligned} 5\ddot{x} &= -5kv \\ \ddot{x} &= -kv \end{aligned}$$

For (b):

$$\begin{aligned} 5\ddot{x} &= -5kv^2 \\ \ddot{x} &= -kv^2 \end{aligned}$$

For (c):

$$\begin{aligned} 5\ddot{x} &= -5kv - 5 \\ \ddot{x} &= -kv - 1 \end{aligned}$$

For (d):

$$\begin{aligned}5\ddot{x} &= -5kv - 5kv^2 - 100 \\ \ddot{x} &= -kv - kv^2 - 20\end{aligned}$$

Question 3:

We use the relation $\ddot{x} = \frac{dv}{dt}$. We separate variables and integrate. Initial condition: $v(0) = 10$.

For (a) $\ddot{x} = v$:

$$\begin{aligned}\frac{dv}{dt} &= v \\ \int \frac{1}{v} dv &= \int dt \\ \ln|v| &= t + C \\ \text{At } t = 0, v = 10 &\implies \ln(10) = C \\ \ln(v) &= t + \ln(10) \\ \ln\left(\frac{v}{10}\right) &= t\end{aligned}$$

For (b) $\ddot{x} = v^2$:

$$\begin{aligned}\frac{dv}{dt} &= v^2 \\ \int v^{-2} dv &= \int dt \\ -\frac{1}{v} &= t + C \\ \text{At } t = 0, v = 10 &\implies -\frac{1}{10} = C \\ -\frac{1}{v} &= t - \frac{1}{10}\end{aligned}$$

For (c) $\ddot{x} = 1 + v$:

$$\begin{aligned}\frac{dv}{dt} &= 1 + v \\ \int \frac{1}{1+v} dv &= \int dt \\ \ln|1+v| &= t + C \\ \text{At } t = 0, v = 10 &\implies \ln(11) = C \\ \ln\left(\frac{1+v}{11}\right) &= t\end{aligned}$$

For (d) $\ddot{x} = 1 + v^2$:

$$\begin{aligned}\frac{dv}{dt} &= 1 + v^2 \\ \int \frac{1}{1 + v^2} dv &= \int dt \\ \tan^{-1}(v) &= t + C \\ \text{At } t = 0, v = 10 &\implies \tan^{-1}(10) = C \\ \tan^{-1}(v) - \tan^{-1}(10) &= t\end{aligned}$$

Question 4:

We rearrange the expressions from Question 3 to solve for v in terms of t .

For (a):

$$\begin{aligned}\ln\left(\frac{v}{10}\right) &= t \\ \frac{v}{10} &= e^t \\ v &= 10e^t\end{aligned}$$

For (b):

$$\begin{aligned}-\frac{1}{v} &= t - \frac{1}{10} \\ \frac{1}{v} &= \frac{1}{10} - t \\ \frac{1}{v} &= \frac{1 - 10t}{10} \\ v &= \frac{10}{1 - 10t}\end{aligned}$$

For (c):

$$\begin{aligned}\ln\left(\frac{1+v}{11}\right) &= t \\ \frac{1+v}{11} &= e^t \\ 1+v &= 11e^t \\ v &= 11e^t - 1\end{aligned}$$

For (d):

$$\begin{aligned}\tan^{-1}(v) - \tan^{-1}(10) &= t \\ \tan^{-1}(v) &= t + \tan^{-1}(10) \\ v &= \tan(t + \tan^{-1}(10))\end{aligned}$$

Question 5:

We use the relation $\ddot{x} = v \frac{dv}{dx}$. Initial conditions: $x(0) = 0, v(0) = 10$.

For (a) $\ddot{x} = 2v$:

$$v \frac{dv}{dx} = 2v$$

$$\frac{dv}{dx} = 2$$

$$\int dv = \int 2dx$$

$$v = 2x + C$$

$$\text{At } x = 0, v = 10 \implies 10 = 0 + C$$

$$v = 2x + 10$$

Rearranging for x :

$$2x = v - 10$$

$$x = \frac{v - 10}{2}$$

For (b) $\ddot{x} = 4v^2$:

$$v \frac{dv}{dx} = 4v^2$$

$$\frac{dv}{dx} = 4v$$

$$\int \frac{1}{v} dv = \int 4dx$$

$$\ln |v| = 4x + C$$

$$\text{At } x = 0, v = 10 \implies \ln(10) = C$$

$$\ln \left(\frac{v}{10} \right) = 4x$$

Rearranging for x :

$$x = \frac{1}{4} \ln \left(\frac{v}{10} \right)$$

For (c) $\ddot{x} = 1 + 3v$:

$$v \frac{dv}{dx} = 1 + 3v$$

$$\int \frac{v}{1 + 3v} dv = \int dx$$

Using substitution $u = 1 + 3v, dv = \frac{1}{3}du, v = \frac{u - 1}{3}$:

$$\int \frac{(u - 1)/3}{u} \frac{1}{3} du = x + C$$

$$\frac{1}{9} \int \left(1 - \frac{1}{u}\right) du = x + C$$

$$\frac{1}{9} (u - \ln |u|) = x + C$$

$$\frac{1}{9} (1 + 3v - \ln |1 + 3v|) = x + C$$

$$\text{At } x = 0, v = 10 \implies C = \frac{1}{9} (31 - \ln 31)$$

Rearranging for x :

$$x = \frac{1}{9} (1 + 3v - \ln |1 + 3v|) - \frac{1}{9} (31 - \ln 31)$$

For (d) $\ddot{x} = 1 + 9v^2$:

$$v \frac{dv}{dx} = 1 + 9v^2$$

$$\int \frac{v}{1 + 9v^2} dv = \int dx$$

Let $u = 1 + 9v^2, du = 18v dv$:

$$\frac{1}{18} \int \frac{1}{u} du = x + C$$

$$\frac{1}{18} \ln |1 + 9v^2| = x + C$$

$$\text{At } x = 0, v = 10 \implies C = \frac{1}{18} \ln(901)$$

Rearranging for x :

$$x = \frac{1}{18} \ln \left(\frac{1 + 9v^2}{901} \right)$$

Question 6:

We rearrange the expressions from Question 5 to find $v(x)$.

For (a):

$$x = \frac{v - 10}{2}$$

$$2x = v - 10$$

$$v = 2x + 10$$

For (b):

$$, x = \frac{1}{4} \ln \left(\frac{v}{10} \right) \quad (1)$$

$$4x = \ln \left(\frac{v}{10} \right) \quad (2)$$

$$\frac{v}{10} = e^{4x} \quad (3)$$

$$v = 10e^{4x} \quad (4)$$

$$(5)$$

For (c):

$$x = \frac{1}{9}(1 + 3v - \ln |1 + 3v|) - \frac{1}{9}(31 - \ln 31)$$

This is a transcendental equation and cannot be solved for v explicitly in terms of elementary functions.

For (d):

$$x = \frac{1}{18} \ln \left(\frac{1 + 9v^2}{901} \right)$$

$$18x = \ln \left(\frac{1 + 9v^2}{901} \right)$$

$$\frac{1 + 9v^2}{901} = e^{18x}$$

$$1 + 9v^2 = 901e^{18x}$$

$$9v^2 = 901e^{18x} - 1$$

$$v = \sqrt{\frac{901e^{18x} - 1}{9}}$$

Skill-building

Question 1:

Given: mass m , initial speed u , resistive force $F = -m(1 + v)$.

(a) Show $a = -(1 + v)$:

$$F = ma$$

$$-m(1 + v) = ma$$

$$a = -(1 + v)$$

(b) Find x in terms of v :

$$\begin{aligned}a &= v \frac{dv}{dx} \\ -(1+v) &= v \frac{dv}{dx} \\ \int dx &= \int \frac{-v}{1+v} dv \\ \int dx &= \int \left(-1 + \frac{1}{1+v} \right) dv \\ x &= -v + \ln|1+v| + C \\ \text{At } x=0, v=u &\implies 0 = -u + \ln(1+u) + C \\ C &= u - \ln(1+u) \\ x &= -v + \ln(1+v) + u - \ln(1+u) \\ x &= (u-v) + \ln\left(\frac{1+v}{1+u}\right)\end{aligned}$$

(c) Find v in terms of t :

$$\begin{aligned}a &= \frac{dv}{dt} \\ \frac{dv}{dt} &= -(1+v) \\ \int \frac{1}{1+v} dv &= \int -dt \\ \ln(1+v) &= -t + C \\ \text{At } t=0, v=u &\implies \ln(1+u) = C \\ \ln(1+v) &= -t + \ln(1+u) \\ \ln\left(\frac{1+v}{1+u}\right) &= -t \\ \frac{1+v}{1+u} &= e^{-t} \\ v &= (1+u)e^{-t} - 1\end{aligned}$$

(d) Find x in terms of t :

$$\begin{aligned}x &= \int v dt \\ x &= \int ((1+u)e^{-t} - 1) dt \\ x &= -(1+u)e^{-t} - t + C \\ \text{At } t=0, x=0 &\implies 0 = -(1+u) - 0 + C \\ C &= 1+u \\ x &= (1+u)(1-e^{-t}) - t\end{aligned}$$

(e) Show $x + v + t = u$:

$$\begin{aligned}x + v + t &= [(1 + u)(1 - e^{-t}) - t] + [(1 + u)e^{-t} - 1] + t \\x + v + t &= (1 + u) - (1 + u)e^{-t} - t + (1 + u)e^{-t} - 1 + t \\x + v + t &= 1 + u - 1 \\x + v + t &= u\end{aligned}$$

(f) Distance and time to come to rest ($v = 0$):

Time t :

$$\begin{aligned}0 &= (1 + u)e^{-t} - 1 \\e^{-t} &= \frac{1}{1 + u} \\-t &= \ln\left(\frac{1}{1 + u}\right) \\t &= \ln(1 + u)\end{aligned}$$

Distance x :

$$x = u - v - t$$

At $v = 0$:

$$\begin{aligned}x &= u - 0 - \ln(1 + u) \\x &= u - \ln(1 + u)\end{aligned}$$

Question 2:

Given: $m = 3$ kg, driving force $F_d = 5$ N, resistance $F_r = 3 + 3v$ N.

(a) Show $\ddot{x} = 1 - v$:

$$\begin{aligned}F_{net} &= ma \\6 - (3 + 3v) &= 3\ddot{x} \\3 - 3v &= 3\ddot{x} \\\ddot{x} &= 1 - v\end{aligned}$$

(b) Show $v = 1 - e^{-t} + v_0e^{-t}$:

$$\begin{aligned}\frac{dv}{dt} &= 1 - v \\\int \frac{1}{1 - v} dv &= \int dt \\-\ln|1 - v| &= t + C \\\ln|1 - v| &= -t - C \\1 - v &= Ae^{-t} \\v &= 1 - Ae^{-t}\end{aligned}$$

$$\text{At } t = 0, v = v_0 \implies v_0 = 1 - A \implies A = 1 - v_0$$

$$\begin{aligned}v &= 1 - (1 - v_0)e^{-t} \\v &= 1 - e^{-t} + v_0e^{-t}\end{aligned}$$

(c) Terminal velocity v_T :

Terminal velocity occurs when $a = 0$:

$$0 = 1 - v$$

$$v_T = 1 \text{ m/s}$$

(d) Distance x when velocity increases from v_0 to v_1 :

$$a = v \frac{dv}{dx}$$

$$1 - v = v \frac{dv}{dx}$$

$$\int dx = \int \frac{v}{1 - v} dv$$

$$\int dx = \int \frac{-(1 - v) + 1}{1 - v} dv$$

$$\int dx = \int \left(-1 + \frac{1}{1 - v}\right) dv$$

$$x = -v - \ln |1 - v| + C$$

From v_0 to v_1 :

$$x = [-v_1 - \ln(1 - v_1)] - [-v_0 - \ln(1 - v_0)]$$

$$x = v_0 - v_1 + \ln(1 - v_0) - \ln(1 - v_1)$$

$$x = (v_0 - v_1) + \ln \left(\frac{1 - v_0}{1 - v_1} \right)$$

Question 3:

Given: $m = 2 \text{ kg}$, $u = 20 \text{ m/s}$, $F_r = 0.5v^2 \text{ N}$.

(a) Show $\frac{dv}{dt} = -0.25v^2$:

$$F = ma$$

$$-0.5v^2 = 2 \frac{dv}{dt}$$

$$\frac{dv}{dt} = -0.25v^2$$

(b) Show $v = \frac{20}{1+5t}$:

$$\begin{aligned}\frac{dv}{v^2} &= -0.25dt \\ \int v^{-2}dv &= \int -0.25dt \\ -\frac{1}{v} &= -0.25t + C \\ \text{At } t = 0, v = 20 &\implies -\frac{1}{20} = C \\ -\frac{1}{v} &= -0.25t - \frac{1}{20} \\ \frac{1}{v} &= 0.25t + 0.05 \\ \frac{1}{v} &= \frac{5t + 1}{20} \\ v &= \frac{20}{1 + 5t}\end{aligned}$$

(c) Displacement x after 2 seconds:

$$\begin{aligned}x &= \int_0^2 v dt \\ x &= \int_0^2 \frac{20}{1 + 5t} dt \\ x &= 20 \left[\frac{1}{5} \ln |1 + 5t| \right]_0^2 \\ x &= 4[\ln(11) - \ln(1)] \\ x &= 4 \ln(11) \approx 9.59 \text{ m}\end{aligned}$$

Question 4:

Given: $m = 1 \text{ kg}$, $u = 1 \text{ m/s}$, $F_r = (1 + v^2) \text{ N}$.

(a) Show $t = \frac{\pi}{4} - \tan^{-1}(v)$:

$$\begin{aligned}F &= ma \\ -(1 + v^2) &= 1 \frac{dv}{dt} \\ \frac{dv}{1 + v^2} &= -dt \\ \int \frac{1}{1 + v^2} dv &= \int -dt \\ \tan^{-1}(v) &= -t + C \\ \text{At } t = 0, v = 1 &\implies \tan^{-1}(1) = C \implies C = \frac{\pi}{4} \\ \tan^{-1}(v) &= -t + \frac{\pi}{4} \\ t &= \frac{\pi}{4} - \tan^{-1}(v)\end{aligned}$$

(b) Time to come to rest ($v = 0$):

$$\begin{aligned}t &= \frac{\pi}{4} - \tan^{-1}(0) \\t &= \frac{\pi}{4} - 0 \\t &= \frac{\pi}{4} \approx 0.785 \text{ s}\end{aligned}$$

Question 5:

Given: $m = 1 \text{ kg}$, $u = \sqrt{3} \text{ m/s}$, $F_r = v + v^3 \text{ N}$.

(a) Acceleration $\frac{dv}{dt} = -(v + v^3)$:

$$\begin{aligned}F &= ma \\-(v + v^3) &= 1 \frac{dv}{dt} \\\frac{dv}{dt} &= -(v + v^3)\end{aligned}$$

(b) Show $x = \tan^{-1} \left(\frac{\sqrt{3}-v}{1+v\sqrt{3}} \right)$:

$$\begin{aligned}v \frac{dv}{dx} &= -(v + v^3) \\v \frac{dv}{dx} &= -v(1 + v^2) \\\frac{dv}{dx} &= -(1 + v^2) \\\int dx &= \int \frac{-1}{1 + v^2} dv \\x &= -\tan^{-1}(v) + C\end{aligned}$$

$$\text{At } x = 0, v = \sqrt{3} \implies 0 = -\tan^{-1}(\sqrt{3}) + C$$

$$C = \frac{\pi}{3}$$

$$x = \frac{\pi}{3} - \tan^{-1}(v)$$

Using the identity $\tan^{-1} A - \tan^{-1} B = \tan^{-1} \left(\frac{A - B}{1 + AB} \right)$:

$$x = \tan^{-1}(\sqrt{3}) - \tan^{-1}(v) = \tan^{-1} \left(\frac{\sqrt{3} - v}{1 + v\sqrt{3}} \right)$$

(c) Show $t = \frac{1}{2} \ln \left(\frac{3(1+v^2)}{4v^2} \right)$:

$$\frac{dv}{dt} = -v(1+v^2)$$

$$\int dt = \int \frac{-1}{v(1+v^2)} dv$$

Using partial fractions: $\frac{1}{v(1+v^2)} = \frac{1}{v} - \frac{v}{1+v^2}$

$$t = - \int \left(\frac{1}{v} - \frac{v}{1+v^2} \right) dv$$

$$t = -[\ln |v| - \frac{1}{2} \ln(1+v^2)] + C$$

$$t = \frac{1}{2} \ln(1+v^2) - \ln |v| + C$$

$$t = \ln \left(\frac{\sqrt{1+v^2}}{v} \right) + C$$

$$\text{At } t = 0, v = \sqrt{3} \implies 0 = \ln \left(\frac{\sqrt{1+3}}{\sqrt{3}} \right) + C$$

$$C = -\ln \left(\frac{2}{\sqrt{3}} \right)$$

$$t = \ln \left(\frac{\sqrt{1+v^2}}{v} \cdot \frac{\sqrt{3}}{2} \right)$$

$$t = \ln \left(\frac{\sqrt{3(1+v^2)}}{2v} \right)$$

$$t = \frac{1}{2} \ln \left(\frac{3(1+v^2)}{4v^2} \right)$$

(d) V^2 as a function of t and limiting x :

$$2t = \ln \left(\frac{3(1+v^2)}{4v^2} \right)$$

$$e^{2t} = \frac{3}{4} \left(\frac{1}{v^2} + 1 \right)$$

$$\frac{4}{3} e^{2t} - 1 = \frac{1}{v^2}$$

$$v^2 = \frac{1}{\frac{4}{3} e^{2t} - 1} = \frac{3}{4e^{2t} - 3}$$

Limiting value of x as $v \rightarrow 0$:

$$x = \frac{\pi}{3} - \tan^{-1}(v)$$

As $v \rightarrow 0, \tan^{-1}(v) \rightarrow 0$:

$$x_{max} = \frac{\pi}{3}$$

Easier Exam Questions

Question 1:

Given: $m = 5 \text{ kg}$, $\theta = 60^\circ$, $g = 10 \text{ m/s}^2$.

(i) Resultant force down the slope:

$$F_{gravity,\parallel} = mg \sin(60^\circ)$$

$$F_{gravity,\parallel} = 5(g) \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}g$$

$$F_{resistive} = 2v + 2v^2$$

$$F_{net} = F_{gravity,\parallel} - F_{resistive}$$

$$F_{net} = \frac{5\sqrt{3}}{2}g - 2v - 2v^2$$

(ii) Constant speed v when $F_{net} = 0$ and $g = 10$:

$$0 = \frac{5\sqrt{3}}{2}(10) - 2v - 2v^2$$

$$0 = 25\sqrt{3} - 2v - 2v^2$$

$$2v^2 + 2v - 25\sqrt{3} = 0$$

$$v = \frac{-2 \pm \sqrt{2^2 - 4(2)(-25\sqrt{3})}}{2(2)}$$

$$v = \frac{-2 \pm \sqrt{4 + 200\sqrt{3}}}{4}$$

$$v = \frac{-2 \pm \sqrt{350.41}}{4}$$

$$v = \frac{-2 \pm 18.72}{4}$$

$$v = 4.18 \text{ or } -5.18$$

Since $v > 0$, $v \approx 4.2 \text{ m/s}$

Question 2:

Given: $m = 50 \text{ kg}$, $\theta = 30^\circ$, $F_{res} = R\sqrt{2} + 6v$.

Resolving forces along the slope:

$$F_{net} = ma$$

$$F - F_{res} + mg \sin(30^\circ) = ma$$

$$F - (R\sqrt{2} + 6v) + 50g(0.5) = 50a$$

$$F - R\sqrt{2} - 6v + 25g = 50a$$

Rearranging for F :

$$F = 50a - 25g + R\sqrt{2} + 6v$$

Comparing with options:

Option (A) matches exactly.

Question 3:

Given: M kg, $v_{top} = 360$ km/h = 100 m/s. Forces: $F_{brake} = \frac{2}{5}M$, $F_{parachute} = \frac{Mv^2}{200}$.

Equation of motion:

$$\begin{aligned}F_{net} &= Ma \\ -\frac{2}{5}M - \frac{Mv^2}{200} &= M \frac{dv}{dx} v \\ -\frac{2}{5} - \frac{v^2}{200} &= v \frac{dv}{dx}\end{aligned}$$

Separating variables:

$$\begin{aligned}dx &= \frac{v}{-\frac{2}{5} - \frac{v^2}{200}} dv \\ dx &= \frac{-200v}{\frac{80+v^2}{200} \cdot 200} dv \\ dx &= \frac{-200v}{80+v^2} dv\end{aligned}$$

Integrating from $x = 0, v = 100$ to x, v :

$$\begin{aligned}x &= \int_{100}^v \frac{-200v}{80+v^2} dv \\ x &= -100 \int_{100}^v \frac{2v}{80+v^2} dv \\ x &= -100 [\ln(80+v^2)]_{100}^v \\ x &= -100 [\ln(80+v^2) - \ln(80+100^2)] \\ x &= 100 \ln \left(\frac{80+100^2}{80+v^2} \right)\end{aligned}$$

Question 4:

Given: $\frac{dv}{dt} = kv$, $v(0) = 6$, $v(10) = 2$.

Solving for $v(t)$:

$$\begin{aligned}\int \frac{1}{v} dv &= \int k dt \\ \ln v &= kt + C \\ v &= Ae^{kt} \\ \text{At } t = 0, v = 6 &\implies A = 6 \\ v &= 6e^{kt}\end{aligned}$$

Finding k using $v(10) = 2$:

$$\begin{aligned}2 &= 6e^{10k} \\ \frac{1}{3} &= e^{10k} \\ 10k &= \ln(1/3) \\ k &= \frac{-\ln 3}{10}\end{aligned}$$

Finding $v(5)$:

$$\begin{aligned}v(5) &= 6e^{5k} \\ v(5) &= 6e^{5(\frac{-\ln 3}{10})} \\ v(5) &= 6e^{-0.5 \ln 3} \\ v(5) &= 6(3^{-0.5}) \\ v(5) &= \frac{6}{\sqrt{3}} = 2\sqrt{3} \\ v(5) &\approx 3.46 \text{ m/s}\end{aligned}$$

Comparing with options:

Option (D) is correct.

Harder Exam Questions

Question 1:

Given: $m = 100\text{g} = 0.1\text{kg}$, $u = 20\text{m/s}$, $F_{wind} = 0.5\text{N}$, $F_{air} = 0.05v\text{N}$.

(i) Acceleration $\ddot{x} = -5 - 0.5v$:

$$\begin{aligned}F_{net} &= ma \\ -0.5 - 0.05v &= 0.1\ddot{x} \\ \ddot{x} &= \frac{-0.5 - 0.05v}{0.1} \\ \ddot{x} &= -5 - 0.5v\end{aligned}$$

(ii) Time $t = -2 \ln \left(\frac{10+v}{30} \right)$:

$$\begin{aligned}\frac{dv}{dt} &= -0.5(10+v) \\ \int \frac{1}{10+v} dv &= \int -0.5 dt \\ \ln(10+v) &= -0.5t + C \\ \text{At } t=0, v=20 &\implies \ln(30) = C \\ \ln(10+v) &= -0.5t + \ln(30) \\ 0.5t &= \ln(30) - \ln(10+v) \\ 0.5t &= \ln \left(\frac{30}{10+v} \right) \\ t &= 2 \ln \left(\frac{30}{10+v} \right) = -2 \ln \left(\frac{10+v}{30} \right)\end{aligned}$$

Time to stop ($v=0$):

$$\begin{aligned}t &= -2 \ln \left(\frac{10}{30} \right) \\ t &= -2 \ln(1/3) = 2 \ln 3 \\ t &\approx 2.197 \text{ s}\end{aligned}$$

(iii) Position $x = 60 - 60e^{-0.5t} - 10t$:

$$\begin{aligned}v &= 30e^{-0.5t} - 10 \\ x &= \int (30e^{-0.5t} - 10) dt \\ x &= \frac{30}{-0.5} e^{-0.5t} - 10t + C \\ x &= -60e^{-0.5t} - 10t + C \\ \text{At } t=0, x=0 &\implies 0 = -60 - 0 + C \implies C = 60 \\ x &= 60 - 60e^{-0.5t} - 10t\end{aligned}$$

Distance travelled before stopping:

$$\begin{aligned}\text{At } t &= 2 \ln 3 : \\ x &= 60 - 60e^{-0.5(2 \ln 3)} - 10(2 \ln 3) \\ x &= 60 - 60e^{-\ln 3} - 20 \ln 3 \\ x &= 60 - 60(1/3) - 20 \ln 3 \\ x &= 40 - 20 \ln 3 \\ x &\approx 40 - 21.97 = 18.03 \text{ m}\end{aligned}$$

Question 2:

Given: $m = 4 \text{ kg}$, $F = 16 - 4v^2 \text{ N}$.

(i) Expression for $x(v)$:

$$F = ma$$

$$16 - 4v^2 = 4 \frac{dv}{dx} v$$

$$4(4 - v^2) = 4v \frac{dv}{dx}$$

$$4 - v^2 = v \frac{dv}{dx}$$

$$\int dx = \int \frac{v}{4 - v^2} dv$$

$$\text{Let } u = 4 - v^2, du = -2v dv \implies v dv = -\frac{1}{2} du :$$

$$x = \int \frac{-1/2}{u} du$$

$$x = -\frac{1}{2} \ln |4 - v^2| + C$$

$$\text{At } x = 0, v = 0 \implies 0 = -\frac{1}{2} \ln 4 + C \implies C = \frac{1}{2} \ln 4$$

$$x = \frac{1}{2} \ln 4 - \frac{1}{2} \ln |4 - v^2|$$

$$x = \frac{1}{2} \ln \left(\frac{4}{4 - v^2} \right)$$

(ii) Expression for $v(t)$ and $x(3)$:

$$a = \frac{dv}{dt} = \frac{16 - 4v^2}{4} = 4 - v^2$$

$$\int \frac{1}{4 - v^2} dv = \int dt$$

$$\text{Using partial fractions: } \frac{1}{4 - v^2} = \frac{1}{4} \left(\frac{1}{2 - v} + \frac{1}{2 + v} \right)$$

$$\frac{1}{4} \int \left(\frac{1}{2 - v} + \frac{1}{2 + v} \right) dv = t + C$$

$$\frac{1}{4} [-\ln |2 - v| + \ln |2 + v|] = t + C$$

$$\frac{1}{4} \ln \left(\frac{2 + v}{2 - v} \right) = t + C$$

$$\text{At } t = 0, v = 0 \implies C = 0$$

$$\ln \left(\frac{2 + v}{2 - v} \right) = 4t$$

$$\frac{2 + v}{2 - v} = e^{4t}$$

$$2 + v = 2e^{4t} - ve^{4t}$$

$$v(1 + e^{4t}) = 2(e^{4t} - 1)$$

$$v = \frac{2(e^{4t} - 1)}{e^{4t} + 1} = 2 \tanh(2t)$$

Displacement after 3 seconds:

$$x(3) = \frac{1}{2} \ln \left(\frac{4}{4 - v(3)^2} \right)$$

$$v(3) = 2 \tanh(6) \approx 2$$

$$x(3) \approx \frac{1}{2} \ln \left(\frac{4}{4 - 2^2} \right)$$

Since $v \rightarrow 2$ as $t \rightarrow \infty$, the displacement x grows logarithmically.

Using $v = 2 \tanh(2t)$, we can integrate $v(t)$:

$$x(3) = \int_0^3 2 \tanh(2t) dt = [\ln(\cosh(2t))]_0^3 = \ln(\cosh 6)$$

$$x(3) \approx \ln(201.7) \approx 5.31 \text{ m}$$

Question 3:

Given: $m = 100\text{kg}$, $F_{\text{pull}} = 800 \cos(60^\circ) = 400\text{N}$ (horizontal component), $F_{\text{res}} = 100kv^2\text{N}$.

(i) Normal force N :

$$\sum F_y = 0$$

$$N + 800 \sin(60^\circ) - 100(10) = 0$$

$$N + 800 \frac{\sqrt{3}}{2} - 1000 = 0$$

$$N = 1000 - 400\sqrt{3} \approx 307.2 \text{ N}$$

(ii) Show $\ddot{x} = 4 - kv^2$ and find k :

$$\sum F_x = ma$$

$$400 - 100kv^2 = 100\ddot{x}$$

$$\ddot{x} = 4 - kv^2$$

Limiting velocity $v_L = 20 \text{ m/s} \implies a = 0$:

$$0 = 4 - k(20^2)$$

$$400k = 4$$

$$k = 0.01$$

(iii) Show $x = -\frac{1}{2k} \ln(1 - \frac{kv^2}{4})$:

$$v \frac{dv}{dx} = 4 - kv^2$$

$$\int dx = \int \frac{v}{4 - kv^2} dv$$

$$\text{Let } u = 4 - kv^2, du = -2kv dv \implies v dv = -\frac{1}{2k} du :$$

$$x = \int \frac{-1}{2k} \frac{1}{u} du$$

$$x = -\frac{1}{2k} \ln |4 - kv^2| + C$$

$$\text{At } x = 0, v = 0 \implies 0 = -\frac{1}{2k} \ln 4 + C$$

$$C = \frac{1}{2k} \ln 4$$

$$x = \frac{1}{2k} \ln 4 - \frac{1}{2k} \ln |4 - kv^2|$$

$$x = -\frac{1}{2k} \ln \left(\frac{4 - kv^2}{4} \right)$$

$$x = -\frac{1}{2k} \ln \left(1 - \frac{kv^2}{4} \right)$$

Distance when $v = 10$ m/s:

$$x = -\frac{1}{2(0.01)} \ln \left(1 - \frac{0.01(10^2)}{4} \right)$$

$$x = -50 \ln \left(1 - \frac{1}{4} \right)$$

$$x = -50 \ln(0.75)$$

$$x \approx 12.9 \text{ m}$$

Question 4:

Given: m kg, $F_{res} = P + Qv^2$ N.

(i) Equation of motion:

$$F_{net} = ma$$

$$-(P + Qv^2) = m \frac{dv}{dt}$$

$$a = \frac{dv}{dt} = -\frac{P + Qv^2}{m}$$

(ii) Distance to slow from $\frac{3U}{2}$ to U if $P = 0$:

$$\begin{aligned} v \frac{dv}{dx} &= -\frac{Qv^2}{m} \\ \frac{dv}{dx} &= -\frac{Qv}{m} \\ \int \frac{1}{v} dv &= \int -\frac{Q}{m} dx \\ \ln v &= -\frac{Q}{m}x + C \end{aligned}$$

Integrating from $v = \frac{3U}{2}$ to $v = U$:

$$\begin{aligned} \ln(U) - \ln\left(\frac{3U}{2}\right) &= -\frac{Q}{m}x \\ \ln\left(\frac{U}{3U/2}\right) &= -\frac{Q}{m}x \\ \ln\left(\frac{2}{3}\right) &= -\frac{Q}{m}x \\ x &= \frac{m}{Q} \ln\left(\frac{3}{2}\right) \end{aligned}$$

(iii) Distance to stop from U if $P > 0$:

$$\begin{aligned} v \frac{dv}{dx} &= -\frac{P + Qv^2}{m} \\ \int dx &= \int \frac{-mv}{P + Qv^2} dv \end{aligned}$$

Let $u = P + Qv^2$, $du = 2Qv dv \implies v dv = \frac{1}{2Q} du$:

$$\begin{aligned} x &= \int \frac{-m}{2Q} \frac{1}{u} du \\ x &= -\frac{m}{2Q} \ln |P + Qv^2| + C \end{aligned}$$

Integrating from $v = U$ to $v = 0$:

$$\begin{aligned} D &= \left[-\frac{m}{2Q} \ln(P + Qv^2)\right]_U^0 \\ D &= -\frac{m}{2Q} [\ln P - \ln(P + QU^2)] \\ D &= \frac{m}{2Q} \ln\left(\frac{P + QU^2}{P}\right) \\ D &= \frac{m}{2Q} \ln\left(1 + \frac{Q}{P}U^2\right) \\ \text{Let } \lambda &= \frac{m}{2Q} \text{ and } k = \frac{Q}{P}. \text{ Then:} \\ D &= \lambda \ln(1 + kU^2) \end{aligned}$$

Question 5:

Given: M kg, $F_{fric} = -\frac{1}{5}M$, $F_{thrust} = -\frac{1}{125}Mv^2$.

(i) Touchdown $v_0 = 80$ m/s. For $0 < t \leq 10$:

$$Ma = -\frac{1}{5}M$$

$$a = -\frac{1}{5} = -0.2 \text{ m/s}^2$$

$$v(t) = 80 - 0.2t$$

At $t = 10$:

$$v(10) = 80 - 0.2(10) = 78 \text{ m/s}$$

$$x(10) = \int_0^{10} (80 - 0.2t) dt = [80t - 0.1t^2]_0^{10}$$

$$x(10) = 800 - 10 = 790 \text{ m}$$

(ii) For $t > 10$:

$$Ma = -\frac{1}{5}M - \frac{1}{125}Mv^2$$

$$a = -\frac{1}{5} - \frac{1}{125}v^2$$

$$a = -\frac{25 + v^2}{125}$$

$$\frac{d^2x}{dt^2} = \frac{-1}{125}(25 + v^2)$$

(iii) Show $x = 790 + 62.5[\ln 6109 - \ln(25 + v^2)]$:

$$v \frac{dv}{dx} = -\frac{25 + v^2}{125}$$

$$dx = \frac{-125v}{25 + v^2} dv$$

$$x - 790 = \int_{78}^v \frac{-125v}{25 + v^2} dv$$

$$x - 790 = -62.5 \int_{78}^v \frac{2v}{25 + v^2} dv$$

$$x - 790 = -62.5[\ln(25 + v^2)]_{78}^v$$

$$x - 790 = -62.5[\ln(25 + v^2) - \ln(25 + 78^2)]$$

$$x - 790 = -62.5[\ln(25 + v^2) - \ln(6109)]$$

$$x = 790 + 62.5[\ln 6109 - \ln(25 + v^2)]$$

(iv) Distance to stop ($v = 0$):

$$x_{stop} = 790 + 62.5[\ln 6109 - \ln(25)]$$

$$x_{stop} = 790 + 62.5 \ln \left(\frac{6109}{25} \right)$$

$$x_{stop} = 790 + 62.5 \ln(244.36)$$

$$x_{stop} = 790 + 62.5(5.498)$$

$$x_{stop} \approx 790 + 343.6 \approx 1134 \text{ m}$$

Question 6:

Given: M kg, $F_{fric} = -\frac{1}{4}M$, $F_{thrust} = -\frac{1}{108}Mv^2$.

(i) Equation of motion:

For $0 < t \leq 20$:

$$Ma = -\frac{1}{4}M \implies a = -\frac{1}{4}$$

$$\frac{d^2x}{dt^2} = -\frac{1}{4}$$

For $t > 20$:

$$Ma = -\frac{1}{4}M - \frac{1}{108}Mv^2$$

$$a = -\frac{1}{4} - \frac{1}{108}v^2$$

$$a = -\frac{27 + v^2}{108}$$

$$\frac{d^2x}{dt^2} = -\frac{1}{108}(27 + v^2)$$

(ii) Touchdown $v_0 = 60$ m/s. At $t = 20$:

$$v(20) = 60 - \frac{1}{4}(20) = 60 - 5 = 55 \text{ m/s}$$

$$x(20) = \int_0^{20} (60 - 0.25t) dt = [60t - 0.125t^2]_0^{20}$$

$$x(20) = 1200 - 0.125(400) = 1200 - 50 = 1150 \text{ m}$$

(iii) Show $x = 1150 + 54\{\ln(27 + 55^2) - \ln(27 + v^2)\}$:

$$v \frac{dv}{dx} = -\frac{27 + v^2}{108}$$

$$dx = \frac{-108v}{27 + v^2} dv$$

$$x - 1150 = \int_{55}^v \frac{-108v}{27 + v^2} dv$$

$$x - 1150 = -54 \int_{55}^v \frac{2v}{27 + v^2} dv$$

$$x - 1150 = -54[\ln(27 + v^2)]_{55}^v$$

$$x - 1150 = -54[\ln(27 + v^2) - \ln(27 + 55^2)]$$

$$x = 1150 + 54\{\ln(27 + 55^2) - \ln(27 + v^2)\}$$